

Served knowledge graph (Neo4j)
not rebuilt; queried

Alternative: one query, no mean-merge, policy at read

raw/
same query, or a narrower one
(30 C, deletions only)

conversion/
essential, lethal $\rightarrow$ fitness 0
kept as its OWN entry

aggregation/
group by genotype
every source entry kept:
screen, temperature, allele, σ , p

LabelPolicy (pydantic, hashed)
precedence: Kuzmin > Costanzo 30 C
0 only if no measurement
 p from the chosen entry

label artifact per policy
one value per label per record
+ subsets and splits as now

025 build as it is: 3.2 TB on disk, five copies

raw/
43,819,983 experiments
839 GB

conversion/
essential, lethal $\rightarrow$ fitness 0
43,819,983 experiments
837 GB

deduplication/
same type + same gene set $\rightarrow$ ONE mean
 $f \leftarrow \text{mean}$, $\sigma \leftarrow \text{RMS}$, $p \leftarrow t\text{-test}$
27,040,733 entries
517 GB

lost here: source values,
screen, temperature, allele,
every source p

aggregation/
group by genotype
13,525,071 records
517 GB

processed/
13,525,071 records + label_df
517 GB

index artifacts (already)
S0 / S2 / S3 subsets, pinned R split,
Q split: lists of record indices